Crystalline Solids vs Amorphous Solids

Atomic/Molecular Arrangement

Crystalline Solids

Crystals have a highly ordered and repetitive arrangement of atoms or molecules in a three-dimensional lattice structure.

Amorphous Solids

The arrangement of atoms or molecules is more random and disordered, lacking a well-defined repeating pattern. The absence of long-range order gives amorphous solids a more irregular structure.

Atomic/Molecular Mobility

Crystalline Solids

In crystals, atoms or molecules are arranged in fixed positions within the lattice structure. This arrangement restricts atomic or molecular mobility, and the solid tends to have a rigid structure.

Amorphous Solids

In amorphous solids, atoms or molecules are not confined to specific lattice positions. They have more freedom to move or vibrate within the material.

Melting Behavior

Crystalline Solids

Crystals have a distinct and well-defined melting point.

Amorphous Solids

Amorphous solids do not have a sharp melting point.

Optical Properties

Crystalline Solids

Crystals typically exhibit well-defined and characteristic optical properties due to their regular and periodic arrangement of atoms or molecules.

Amorphous Solids

Amorphous solids have more random and disordered atomic or molecular arrangements, resulting in a lack of well-defined optical properties.

Mechanical Properties

Crystalline Solids

Crystals tend to have anisotropic mechanical properties, meaning their mechanical behavior varies with different crystallographic directions.

Amorphous Solids

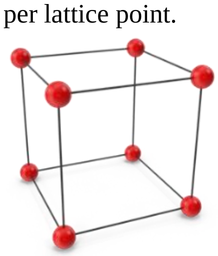
Amorphous solids often exhibit isotropic mechanical properties, meaning their mechanical behavior is relatively uniform in all directions.

Types of Crystal Lattice Structures

Simple Cubic, Face-Centered Cubic, and Body-Centered Cubic are three common crystal lattice structures found in metals.

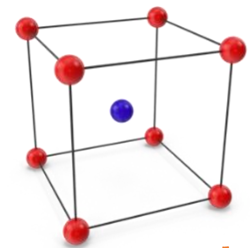
Simple Cubic

In a simple cubic lattice, the atoms or ions occupy the corners of a cube, with one atom per lattice point.



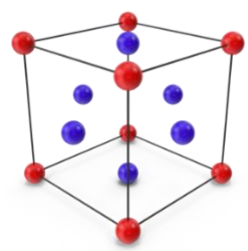
Body-Centered Cubic

In a body-centered cubic lattice, the atoms or ions occupy the corners of the cube as well as one atom at the center of the cube.



Face-Centered Cubic

In a face-centered cubic lattice, the atoms or ions occupy the corners and the centers of each face of the cube.



	Number of atoms	Coordination Number
Simple Cubic	1	6
Body-Centered Cubic	2	8
Face-Centered Cubic	4	12

	Atomic Packing Factor (apf)
Simple Cubic	52.4%
Body-Centered Cubic	68%
Face-Centered Cubic	74%